

RIDA ARSHAD

Final Year Student, Ballari Institute of Technology and Management (BITM)

(+91) 9955339479 | ridaarshad6577@gmail.com | [LinkedIn](#) | [GitHub](#)

EDUCATION

Ballari Institute of Technology and Management

B.E. in Computer Science and Engineering (Data Science)

CGPA: 9.66

2023 – 2027

Ballari, India

TECHNICAL SKILLS

Programming: Python, R, MySQL

Libraries/Frameworks: Pandas, NumPy, Matplotlib, Seaborn, OpenCV, YOLOv8, Flask, Streamlit, Plotly, MongoDB

Data Science Skills: Exploratory Data Analysis (EDA), Machine Learning, Data Visualization & Interpretation, Data Cleaning & Validation, Data Quality Checks (DQC), KPI Engineering, Model Training & Evaluation, Time-Series Analysis, Root Cause Trend Analysis

Tools: Jupyter Notebook, VS Code, Git, MS Excel

WORK EXPERIENCE

Intern – JSW Steel Ltd., Vijayanagar Works [🔗](#)

Nov 2025 – Jan 2026

Tech: Streamlit, Python, MongoDB, SQLite, MS Excel

- Built an industrial-grade **NCO Quality Analytics Dashboard** using Streamlit and Python with **multi-level filtering** (unit, product, defect, financial year, date range), **dynamic product–unit dependency**, and **unit comparison support**.
- Designed **financial-year based defect analysis** with **automated NCO% calculation** using production-weight logic, **multi-index monthly tables** (Total Weight | NCO Weight | NCO%), and **conditional heatmap visualization** for rapid risk detection.
- Implemented statistical quality tools including **Pareto analysis**, **SPC control charts (Cp/Cpk)**, **unit performance ranking**, and **exportable reports (CSV/PDF)** to support manufacturing process improvement.

PATENTS & PUBLICATIONS

Patent Application (Published, Grant-Pending)

Dec 2025

An Automated Stereo Vision-Based Framework for Real-Time Pothole Detection, Severity Assessment, and Optimized Maintenance

Indian Patent Office, Application No.: 202541114081 A

- **Technologies:** YOLOv8, OpenCV, Python, NumPy, Streamlit, GPS-based logging.
- **Domain:** Computer Vision, Intelligent Transportation Systems.

Research Paper: [🔗](#)

PotBot: An AI-Powered Pothole Detection And Road-Damage Prioritization System

2026 International Conference on Emerging Technologies and Future Innovations (ETFI) — IEEE Pune Section

Publisher: IEEE

Apr 2026

DOI: [10.1109/ETFI68128.2026.11484668](https://doi.org/10.1109/ETFI68128.2026.11484668)

PROJECTS

PotBot – AI-Powered Pothole Detection System

Apr 2025 – Present

Tech: YOLOv8, OpenCV, Python, NumPy, Pandas, FastAPI, MongoDB, Stereo Vision

- Trained and fine-tuned **YOLOv8 models on annotated road-condition datasets**; **optimized confidence thresholds** and non-max suppression for **real-time inference on dashcam** video streams.
- Deploying **stereo-vision depth estimation** using disparity mapping to compute pothole depth and surface area for **multi-class severity labeling**.
- Integrating **GPS-based geotagging pipeline** to associate detections with **latitude–longitude coordinates** and enable route-level aggregation of road damage events.
- Designed RESTful backend services using FastAPI for ingestion of detection metadata including **bounding boxes**, **severity scores**, **timestamps**, and **frame samples**.

- Modeled and implemented **MongoDB schemas for structured storage of detection logs and efficient querying by location, severity, and time range.**
- Built **data preprocessing and analytics pipelines** using NumPy and Pandas to generate structured datasets for evaluation, reporting, and visualization.
- Integrated modular pipeline covering frame preprocessing, YOLO inference, depth computation, GPS tagging, and backend logging for **end-to-end automation.**
- **GitHub:** [PotBot](#)

Banking Marketing Campaign Analysis

Feb 2025 – Apr 2025

Tech: Python, Pandas, NumPy, Matplotlib, Seaborn

- Performed **exploratory data analysis** on a **real-world banking dataset** comprising **45,211 records** and **18 features** to study **drivers of term deposit subscription behavior.**
- Executed **data cleaning and preprocessing** including missing value treatment, categorical encoding, outlier inspection, and type normalization for numerical and categorical attributes.
- Conducted univariate and multivariate **statistical analysis** on demographic, financial, and campaign-related variables to quantify their relationship with the binary target variable.
- Computed **correlation matrices** and **visualized feature interactions** using heatmaps to identify **high-impact predictors** such as call duration, account balance, and previous campaign outcome.
- Analyzed **temporal patterns across months and call durations** to detect seasonality and response rate variation using grouped aggregations and distribution plots.
- Evaluated campaign efficiency metrics including contact frequency, conversion rates, and diminishing returns across repeated client interactions.
- Designed **15+ analytical visualizations** (histograms, count plots, box plots, pie charts) to communicate customer segmentation and behavioral trends.
- Generated **actionable insights to improve targeting strategies** by prioritizing high-balance clients, optimal call timing, and historically responsive segments.
- Report: [Finlatics](#) | GitHub: [Finlatics](#)

RailGuard – Dynamic Railway Traffic Simulation & Safety System

Sept 2025 – Present

Tech: Python, Flask, NetworkX, Threading, HTML, CSS, JavaScript

- Implemented a **discrete-time train motion simulator** with 1-second ticks using **kinematic equations** for **velocity update, acceleration control, and position integration.**
- Computed **braking distance** using $d = \frac{v^2}{2a}$ and enforced **dynamic safety envelopes** by maintaining **adaptive headway** with an **additional fixed 200 m buffer.**
- Modeled railway topology as a **weighted directed graph** in NetworkX and applied **A* pathfinding** with **dynamic edge weight inflation** to **avoid occupied track segments.**
- Designed **moving-block signalling logic** by **recalculating safe-speed constraints per train** based on braking envelope intersection and downstream occupancy.
- Implemented **multi-threaded state updates** with **mutex locks** to **ensure consistency** of train positions, velocities, and block occupancy during concurrent simulation steps.
- Developed RESTful control endpoints in Flask for train dispatch, state retrieval, and **simulation reset with JSON-based state serialization.**
- Built a **real-time visualization layer** in JavaScript to render train coordinates, velocity vectors, planned routes, and safety boundaries on a browser-based interface.
- Structured simulation pipeline into modular components for routing, motion update, braking validation, block control, and UI synchronization.
- **GitHub:** [RailGuard](#)

CERTIFICATIONS

Gemini Certified Student(University) — Google for Education
 Python for Data Science, AI and Development — IBM via Coursera
 Internship Training in Python — EZ Trainings
 Data Analysis with R — Google via Coursera
 Advanced Excel — Internshala
 AWS Cloud Quest Practitioner — AWS Skill Builder
 Data Science with Python — Finlatics Data Science Program

ACHIEVEMENTS

Rank 1 — CSE (Data Science) — Year 2023–24
 Rank 2 — CSE (Data Science) — Year 2024–25
 1st Place — Spark Tank (Ideathon)
 1st Place — Debate Competition
 Letter of Recommendation — Finlatics Program
[Refer](#)